

PERFORMANCE TESTING

Project Name

ComicCraft – AI Powered Comic Story Generator

Team ID: _____

Test Date: _____

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing and Stress Testing
Target Module / API	Comic Generation API, Story Generation API, Image Generation Workflow, PDF Generation API
Test Environment	Local development server / Cloud-hosted server
Test Date	_____
Application Backend	Python FastAPI
Primary Test Objective	Evaluate the response time, throughput, error rate, CPU utilization, and memory utilization of ComicCraft under expected and increased user load.

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Test the ComicCraft homepage and basic API availability under normal user traffic.	10	60	Application responds successfully without significant errors.
2	Test the story-generation API with multiple users submitting comic prompts simultaneously.	25	120	Story-generation requests are processed successfully with acceptable response time.
3	Test the complete comic-generation workflow including story generation and panel processing.	50	180	System continues processing requests without crashes or excessive failures.
4	Stress test the comic-generation API with increased concurrent requests.	100	300	System remains stable and gracefully handles increased load.

Step 3: Performance Test Results

Performance Metrics

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Average)	< 2 seconds	To be measured	Pending	Measure using JMeter response-time statistics.
2	Response Time (Maximum)	< 5 seconds	To be measured	Pending	Verify maximum API response time under test load.
3	Throughput (Req/sec)	As high as possible while maintaining stability	To be measured	Pending	Record requests successfully processed per second.
4	Error Rate	< 1%	To be measured	Pending	Monitor failed API requests and generation failures.
5	CPU Utilization	< 80%	To be measured	Pending	Monitor backend server CPU consumption.
6	Memory Utilization	< 80%	To be measured	Pending	Monitor RAM consumption during concurrent requests.

Performance Acceptance Criteria

ComicCraft will be considered performance-ready when the following target conditions are achieved:

- Average response time remains below 2 seconds for normal API operations.
- Maximum response time remains below 5 seconds for normal backend requests.
- Error rate remains below 1%.
- CPU utilization remains below 80% during expected load.
- Memory utilization remains below 80% during expected load.
- The application does not crash during the defined load tests.
- The backend can process multiple concurrent requests without significant degradation.

Note: AI image generation and complete comic generation may naturally take longer than ordinary API operations because they depend on external AI services. Therefore, response-time targets should be evaluated separately for standard API operations and end-to-end AI generation workflows.

Step 4: Observations & Analysis

Performance testing of ComicCraft focuses on determining whether the application can handle multiple users requesting comic generation at the same time.

The main performance-sensitive components are the AI story-generation service, AI image-generation service, comic layout processing, and PDF generation.

During normal operation, lightweight API requests such as health checks, prompt validation, and page requests are expected to respond quickly. AI-powered operations may require additional processing time because they involve communication with external Generative AI services.

The load-testing process should identify:

- Average API response time.
- Maximum API response time.
- Number of successful requests.
- Number of failed requests.
- Requests processed per second.
- CPU consumption.
- Memory consumption.
- Performance degradation as concurrent users increase.

If performance decreases significantly as the number of users increases, possible optimization approaches include asynchronous processing, request queues, caching, connection pooling, optimized image processing, and independent scaling of AI-related services.

Performance Test Analysis by Module

1. Frontend

The frontend is evaluated for its ability to load the application interface and submit comic-generation requests efficiently.

Expected result:

- Fast page loading.
 - Responsive user interaction.
 - No unnecessary repeated API requests.
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2. Story Generation API

The story-generation API receives the user's prompt and communicates with the configured LLM service.

Performance factors include:

- Prompt size.
- Number of concurrent requests.
- External LLM response time.
- Backend processing time.

Expected result:

The API should remain stable under normal concurrent usage.

3. Image Generation

Image generation is one of the most computationally and network-intensive stages of ComicCraft.

Performance depends on:

- Image model response time.
- Image resolution.
- Number of panels.
- Concurrent generation requests.
- External API availability.

Expected result:

The system should process image-generation requests reliably and provide appropriate progress/error feedback.

4. Comic Layout Generation

The layout module combines generated images, dialogue, narration, and captions into comic pages.

Performance factors include:

- Number of panels.
- Image resolution.
- Image-processing operations.
- Available CPU and memory.

Expected result:

Comic pages should be generated without excessive resource consumption.

5. PDF Generation

The PDF module converts the completed comic layout into a downloadable document.

Performance factors include:

- Number of pages.
- Number and resolution of images.
- PDF compression.
- Available system resources.

Expected result:

The PDF should be generated successfully without application failure.

Test Result Interpretation

Pass

A test is marked **Pass** when the measured performance meets the defined target and the application remains stable.

Fail

A test is marked **Fail** when the measured value exceeds the defined target or the application experiences unacceptable errors, crashes, or resource usage.

Pending

A test is marked **Pending** when the test has not yet been executed or the actual measurement has not yet been recorded.

Step 5: Screenshots / Evidence

The following evidence should be attached to this section after executing the performance tests:

Screenshot 1 – JMeter Test Plan

Attach a screenshot showing:

- Thread Group
- Number of virtual users
- Ramp-up period
- Test duration

Screenshot 2 – Response Time Results

Attach the JMeter results showing:

- Average response time
- Minimum response time
- Maximum response time
- Number of requests

Screenshot 3 – Throughput

Attach the graph/table showing requests processed per second.

Screenshot 4 – Error Rate

Attach the results showing successful and failed requests.

Screenshot 5 – CPU and Memory Usage

Attach the system monitoring output showing CPU and RAM utilization during testing.

Recommended Evidence Table

Evidence	Screenshot / Report to Attach
Test Configuration	JMeter Thread Group configuration
Load Test Execution	JMeter test execution screen
Response Time	JMeter Summary Report / Aggregate Report
Throughput	JMeter Throughput Graph
Error Rate	JMeter Results / Summary Report
CPU Usage	System monitoring screenshot
Memory Usage	System monitoring screenshot

Overall Performance Testing Conclusion

Performance testing provides an understanding of how ComicCraft behaves under normal and increased workloads. The testing focuses on the application's backend APIs, AI-powered comic-generation workflow, image processing, and PDF generation.

The performance evaluation should verify that ComicCraft can handle concurrent requests while maintaining acceptable response times, low error rates, and reasonable CPU and memory utilization.

The final performance status should be determined using the **actual measured results obtained from the selected testing tool**.